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Androgen treatment and recovery of function following recurrent laryngeal nerve injury in the rat

Todd J. Brown[a,b,1,*], Amy L. Pittman[c,1], Gina N. Monaco[e], Brent J. Benscoter[c], Avinash V. Mantravadi[c], Lee M. Akst[d], Kathryn J. Jones[a,b] and Eileen M. Foecking[c,f]
[a]*R & D Services Richard L. Roudebush VA Medical Center, Indianapolis, IN, USA*
[b]*Department of Anatomy & Cell Biology, Indiana School of Medicine, Indianapolis, IN, USA*
[c]*Departments of Otolaryngology – Head & Neck Surgery, Loyola University Medical Center, Proviso, IL, USA*
[d]*Department of Otolaryngology–Head and Neck Surgery, Johns Hopkins Hospital, Baltimore, MD, USA*
[e]*Cell Biology, Neurobiology, and Anatomy Program Loyola University Chicago, Stritch School of Medicine, Maywood, IL, USA*
[f]*R and D Services Edward Hines Jr. VA Hospital, Hines, IL, USA*

Abstract.
Purpose: To investigate the effects of the androgen testosterone propionate (TP), on regeneration of the recurrent laryngeal nerve (RLN) after unilateral crush injury using assessment of vocal fold mobility (VFM) as a measure of behavioral recovery.
Methods: 48 adult male rats underwent standardized crush injury of left RLN and received treatment in the form of 2 silastic capsules containing TP or controls receiving a blank capsule (untreated). Direct laryngoscopic assessment of vocal cord mobility was performed before, immediately following and 1, 2, 3, 4, 5 or 6 weeks post injury.
Results: Treatment with TP enhanced the recovery of full VFM following crush injury of the RLN compared to controls. There was statistically significant improvement in VFM seen at the 1 and 2 week time points ($p < 0.05$). By 4 weeks TP-treated rats displayed a 100% recovery of VFM function, compared to only 50% by the control group.
Conclusions: TP enhances RLN functional recovery following a crush injury, which further supports its potential general applicability as a therapeutic agent in peripheral nerve injury.

Keywords: Testosterone propionate, recurrent laryngeal nerve, vocal fold mobility, crush injury

1. Introduction

Previous work in our lab and others have demonstrated the efficacy of gonadal steroid treatment in animal models of peripheral nerve injury (Brown et al., 1999, 2001; Hetzler et al., 2008; Jones, 1993; Jones & Oblinger, 1994, 1989, 1991; Kujawa et al., 1993). Since the majority of motoneuron populations in the CNS contain androgen receptors (Kujawa et al., 1993; Breedlove & Arnold, 1981; Greco et al., 1998; Matsumoto et al., 1993; Simerly et al., 1990), these earlier studies have been focused on examining the effects of androgens on the regeneration of motoneurons in the brain and spinal cord. Treatment with a long-lasting

[1]Both authors contributed equally to the preparation of this manuscript.

*Corresponding author: Todd J. Brown, Ph.D., Richard L. Roudebush VAMC, 1481W. 10th Street, Mail Code 151, Rm C-3074, Indianapolis, IN 46202, USA. Tel.: +1 317 988 4994; Fax: +1 317 278 2040; E-mail: tojobrow@iupui.edu.

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form of testosterone, testosterone propionate (TP), has been shown to be particularly effective when administered at the time of injury in both males and females (Jones, 1993; Tanzer & Jones, 2004). This treatment strategy works for cranial and spinal motoneurons, both exhibiting an increase in the rate of regeneration, an upregulation of structural genes necessary for axonal repair, and a decrease in the time needed for behavioral recovery (Brown et al., 1999, 2001; Hetzler et al., 2008; Jones, 1993; Jones & Oblinger, 1994; Kujawa et al., 1991; Sharma et al., 2009, 2010a, 2010b). Androgens are in use clinically in both men and women, and given in supraphysiological doses for a number of different disorders including HIV-related weight loss, hypogonadism and female androgen deficiency (Strawford et al., 1999; Giuberti et al., 2005; Arlt, 2006).

Most of our work involving cranial motoneuron injury and repair has been based on the rodent facial nerve injury model: hamster, rat and mouse (Kujawa et al., 1991; Jones et al., 1997; Fargo et al., 2008; Tetzlaff et al., 2007). We were interested in seeing if TP treatment would be as effective in another cranial nerve injury paradigm. If so, this would provide additional evidence suggesting that TP treatment is an effective treatment strategy for generalized peripheral nerve injuries. For this reason, as well as its relevance to clinical literature, the rat recurrent laryngeal nerve (RLN) was chosen.

The vagus is well known as the bodies' major source of parasympathetic innervation to glands as well as smooth and cardiac muscles. However, it also has three major branchial branches which serves to provide special visceral efferents to the skeletal muscles of the tongue, pharynx and larynx. The RLN is one of these branches of the vagus, whose branchimotor neurons are located in the nucleus ambiguus in the medulla, and carries motor fibers that supply all of the intrinsic muscles of the larynx, with the exception of the cricothyroid muscle, stimulating abduction and adduction of the vocal folds for phonation, breathing, and protection of the lungs during swallowing. While the RLN is the major motor nerve supply to the intrinsic laryngeal muscles, the superior laryngeal nerve does provide some motor branches to the thyroartenoid and interarytenoid muscles (Wu et al., 1994; Sanders & Mu, 1998). RLN injury is a known complication of surgery in the neck and thorax. Determining the precise incidence of RLN injury is problematic as it appears to vary with the date as well as geographic location of the study (Myssiorek, 2004). However, a recent study found that iatrogenic surgical injury to the RLN accounts for 46.3% of cases of unilateral vocal fold paralysis (Rosenthal et al., 2007). Recovery following inadvertent injury is variable and depends upon the mode and extent of the initial insult. The RLN is fragile and can be damaged even though it may appear to be patent to the surgeon with damage only revealed upon post-surgical endoscopic examination of the patient. The crush model in this study is used primarily to mimic the instances of temporary immobility seen in cases of inflammation, infection or iatrogenic stretching of the RLN (Rosenthal et al., 2007).

Regeneration of the nerve involves tolerating the initial stress response, upregulating restorative genes, producing proteins, inducing new growth of axons, which still terminates in varying degrees of nerve function (Sharma et al., 2010a; Fargo et al., 2008; Jones et al., 1999). Clinically, a paralysis or weakness of one RLN will cause some degree of incomplete glottic closure, resulting in dysphonia, dysphagia, and in some cases, aspiration (Araki et al., 2005; Tessema et al., 2009). Injury to one or both nerves can be devastating to the patient, and prognosis of functional recovery can be unpredictable and challenging (Tessema et al., 2009). Current treatment options are aimed at surgical medialization of the vocal cord, but not at restoring vocal fold mobility (VFM) (Rubin et al., 2003). Surgical treatments have evolved, but outcomes are limited by laryngeal muscle atrophy, and lack of dynamic responses of the vocal folds in response to changes in airflow or pressure.

While the benefits of androgens have been demonstrated in other peripheral nerve injury models, there have been no prior studies to date that investigate their role in RLN injury. Androgen administration may be a potential non-invasive, general treatment strategy for peripheral nerve injury. Thus, the purpose of this study was to investigate the effects of the androgen, TP, on regeneration of the RLN after unilateral crush injury.

2. Methods

2.1. Animals

Adult male Sprague-Dawley rats (200–250 gm) were housed in pairs and given food and water ad libitum. All animals were castrated under isofluorane anesthesia and allowed to recover for 3–5 days.

This time period allowed endogenous androgen to be removed in order to standardize all animal hormonal paradigms for scientific reproducibility before undergoing the nerve crush procedure. Subjects were then divided into two groups: untreated control group, RLN crush injury + 2 blank capsules ($n = 28$); TP treatment group, RLN crush injury + 2 TP capsules ($n = 23$). Animals were randomly assigned to survival times of 1, 2, 3, 4, 5 and 6 weeks post-operative. Group size and TP dosage were chosen based on results from our previous studies involving peripheral nerve injury and treatment with gonadal steroids (Brown et al., 1999; Hetzler et al., 2008). All animals received humane care in compliance with the U.S. National Research Council's *Guide for the Care and Use of Laboratory Animals* and all animal work performed in this study was approved by the Institutional Animal Care and Use Committee of the Edward Hines VA Hospital, Hines, IL.

2.1.1. TP capsules

The TP treatment group received steroid hormones by subcutaneously implanting two Silastic capsules (0.062 inch inner diameter id, 0.095 inch outer diameter, 10 mm length) between the scapulae, with each capsule containing 100% crystalline TP (Sigma) as previously described (Hetzler et al., 2008). The capsules were equilibrated in sterile physiological saline and implanted within 10 minutes of the crush injury. These steroid capsules have been shown to produce serum TP levels of 10.36 ± 0.32 ng/ml at 15–19 days post axotomy (Hetzler et al., 2008). Animals in the untreated group were similarly anesthetized and received implants of 2 blank capsules.

2.1.2. Vocal fold observation and RLN crush injury

Animals were anesthetized for nerve crush and vocal fold observation using isofluorane gas (3%). The animals were placed in a stereotactic operating system in the supine position. A silk suture was used to retract the tongue while a 0° Endogo (Medtronic) operating endoscope with an epiglottis elevator was inserted into the larynx to observe and record vocal fold movement during respiration. A specialized Y-adaptor for the tubing was created to allow both the isofluorane gas and the endoscope access to the upper respiratory tract. This enabled the investigator to observe vocal fold mobility with the endoscope while maintaining an appropriate level of anesthesia. A midline incision was made in the skin between the sternal notch and mandible. Subcutaneous fat and glands were retracted, exposing the strap muscles. The left and right sternohyoid muscles were separated by blunt dissection and retracted to expose the underlying larynx and trachea. The RLN was easily identified on the lateral surface of the trachea. Using a pair of fine jeweler's forceps, the left nerve was crushed twice, at 30 seconds/crush. All injuries were performed by the same individual; a method that has, in our hands, produced reliable and consistent peripheral nerve injuries. Evidence of RLN regeneration and behavioral recovery was obtained by observing vocal fold movement during inspiration and expiration via endoscopy. This was performed in anesthetized animals immediately prior to and after RLN crush injury and at the end of the survival time. After surgery and endoscopy were complete, the strap muscles were reapposed and the overlying skin closed with wound clips. The incision was treated with topical antibiotic and was observed daily for any post-surgical complications. Animals tolerated this procedure quite well and there were no complications arising from surgery.

2.1.3. Vocal fold movement observation and analysis

The vocal cord movements were recorded via a portable video capture device which is part of the Endogo scope (Acme Co., Boston, MA). Immediately before and after nerve crush surgeries, unevoked movements of the vocal folds during spontaneous respiration in anesthetized animals were recorded to confirm the absence or presence of unilateral paralysis. They were also obtained at the time of euthanasia (1, 2, 3, 4, 5 or 6 weeks post-crush injury) to assay the extent of functional recovery, if any. Vocal fold movements were graded on an integral scale of 0 to 4. A score of 0 was given for immobilization of the vocal fold with motion at 0–24% range of motion and a score of 4 was given for full and vigorous adductor and abductor movement. A score of 1 was given to vocal folds with severe paresis where a twitch was present and the estimated motion was between 25–49%. A score of 2 was given to animals with moderate vocal fold paresis with an estimated motion between 50–74%. Lastly, a score of 3 was given to vocal folds with moderate paresis estimated to 75–99% movement. Paralysis was defined as the inability of the laryngeal muscles to abduct the affected side of the vocal fold compared to the intact side. Video recordings of vocal fold movement were scored by a panel of at least 3 investigators who were blinded as to control or experimental nature

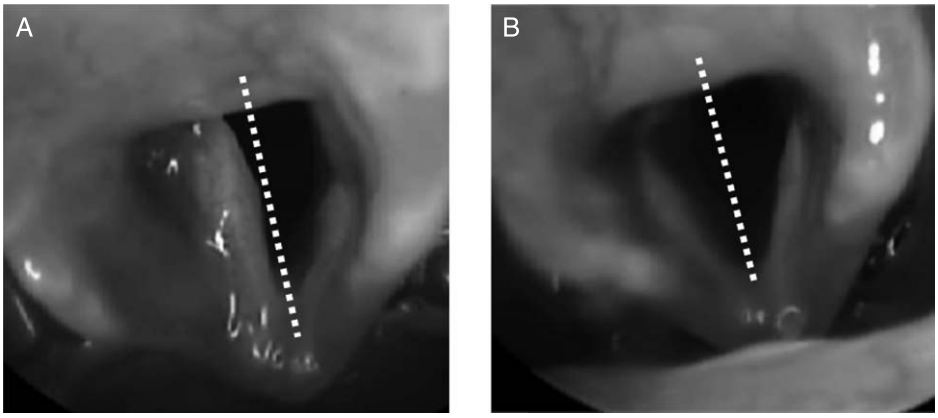


Fig. 1. Endoscopic visualization of vocal fold movment (VFM). **A)** Immobile left vocal fold after crush injury of the ipsilateral RLN. VFM = 0 **B)** Normal movement in an uninjured animal. VFM = 4. Dotted line indicates midline.

of the animal at the time of scoring. Further, inter-rater reliability was tested to ensure the uniformity of scoring.

2.2. Statistical analysis

A 4 × 2 × 2 log-linear analysis of the percentage of animals achieving complete recovery was carried out to examine the association between time, treatment, and recovery status. A Two-way ANOVA (treatment × time) was utilized along with the Holm-Sidak *post-hoc* test to examine the relationship treatment and time on vocal fold mobility scores.

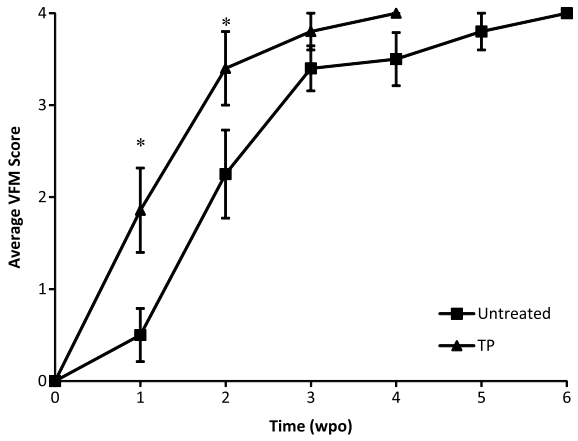


Fig. 2. Effects of Supraphysiological Testosterone Propionate (TP) Treatment on Vocal Fold Mobility (VFM) following a Recurrent Laryngeal Nerve (RLN) crush injury. Shown are the average VFM scores at 1, 2, 3, 4, 5, and 6 weeks after RLN crush injury comparing Untreated and TP-treated animals. Vertical lines represent standard error of the mean. $n = 4$–7/group. Each time point represents a separate cohort of animals. *indicates statistically significant differences between Untreated and TP groups ($p < 0.05$).

3. Results

Endoscopic examination of the larynges of all animals immediately after they had received a crush injury of the left RLN revealed paralysis of the left vocal fold (Fig. 1). A vocal fold mobility (VFM) score of 0, for complete paralysis, was given to all these animals. Animals treated with TP immediately following an RLN crush injury resulted in significantly higher ($p < 0.05$) vocal fold mobility scores at post-operative weeks (wpo) 1 and 2 (Fig. 2). However, as both groups were reaching the ceiling of full recovery by 3 and 4 wpo, the group differences were no longer significant at these time points. The Two-way ANOVA with Holm-Sidak *post-hoc* test revealed a statistically significant effect of treatment [$F(1, 32) = 10.31$, $p < 0.01$] and a statistically significant effect of time [$F(3, 32) = 21.93$, $p < 0.001$]. No statistically significant interaction was observed between time and treatment [$F(3, 32) = 1.08$, n.s.].

The percentage of animals with complete VFM recovery (all animals scoring a 4) following a crush injury was increased in the TP group at 1, 2, 3 and 4 wpo (Table 1). 100% of the TP-treated animals exhibited a full recovery of VFM at 4 wpo, while only 50% of the untreated controls showed full recovery at that time point (Table 1). The time at which all animals in the TP

Table 1
Effects of Supraphysiological Testosterone Propionate (TP) Treatment on Time Required to Complete Functional Recovery after Recurrent Laryngeal Nerve (RLN) crush injury. Shown are the percentages of animals in each group that achieved a vocal fold mobility score of 4 (defined as complete functional recovery) at 1, 2, 3, 4, 5, and 6 weeks after RLN crush injury comparing Untreated and TP-treated animals. Listed below each percentage in parentheses is the number of recovered animals out of the entire set at each particular time point ($n = 4$–7/group). Each time point represents a separate cohort of animals

	Percentage of animals achieving complete functional recovery following a RLN crush					
Treatment group	Weeks post operative					
	1	2	3	4	5	6
Untreated	0 (0/4)	0 (0/4)	40 (2/5)	50 (2/4)	80 (4/5)	100 (6/6)
TP	14 (1/7)	60 (3/5)	80 (4/5)	100 (6/6)		

group exhibited full recovery occurred 2 weeks prior to those in the untreated group. None of the untreated animals achieved complete recovery within the first 2 wpo. The log-linear analysis revealed that the percentage of animals reaching full recovery was dependent on both time and treatment [$G^2(10) = 25.76$, $p < 0.01$]. There were main effects of both time [$G^2(3) = 13.44$, $p < 0.01$] and treatment [$G^2(1) = 5.72$, $p < 0.05$], but there was no significant interaction between time and treatment [$G^2(3) = 0.46$, n.s.]. Untreated animals in that received blank capsules did not achieve full recovery of VFM until 6 wpo. The treatment with TP significantly decreased the time until full recovery of VFM from a RLN crush by two weeks which equates to a 33% reduction in overall recovery time.

4. Discussion

Systemic administration of TP has been demonstrated to increase the rate of axonal regeneration, as well as accelerate behavioral recovery from facial and hindlimb paralysis after injury to the facial and sciatic nerves, respectively (Brown et al., 1999; Kujawa et al., 1993; Sharma et al., 2009; Jones et al., 1999; Jones et al., 2001). Further, this treatment is most effective when administered during the first hours post-injury (Tanzer & Jones, 2004). Androgens have been proposed as a general treatment for injured peripheral nerves and have been used successfully in several nerve injury models (Kujawa et al., 1989; Kujawa et al., 1993; Yu & Yu, 1983). Based on the results of past studies, the aim of the current study was to determine if androgen treatment can accelerate behavioral recovery after injury in another cranial nerve: the recurrent laryngeal nerve.

The RLN model was chosen for its clinical relevance, as injury to the RLN is a common complication of surgery in the neck and chest (Myssiorek, 2004). The RLN is one of the longest branches of the cranial nerves, branching from the descending Vagus nerve at the level of the mediastinum, looping around the aortic arch on the left and the subclavian artery on the right, then traveling superiorly along the tracheo-esophageal groove. It enters the larynx on both sides behind the articulation of the inferior cornu of the thyroid cartilage and the cricoid cartilage, where it supplies sensory fibers to the internal surface of the larynx inferior to the vocal folds and innervates all of the intrinsic muscles of the larynx, except the cricothyroid muscle. To date, treatment regimens include medialization procedures or surgery to alter the anatomy of the larynx (Rubin et al., 2003). Interestingly, there have been no studies to investigate the effect of androgens on RLN functional recovery following injury.

A gradual recovery in VFM after unilateral crush injury of the RLN over a period of 6 weeks in untreated and 4 weeks in treated (TP) animals was observed. With respect to the untreated animals, their recovery followed a time course similar to that observed in other studies utilizing a unilateral crush injury (Araki et al., 2005; Tessema et al., 2009). Tessema et al. noted moderate recovery by 3 weeks post-op (Tessema et al., 2009), with full recovery being observed by 6 weeks post-op (Tessema et al., 2009, 2008). We observed that after crush injury only half of the animals showed recovery 4 weeks post-injury and no recovery was observable at 2 weeks post-op; this is an effect that has been observed on other studies (Araki et al., 2005; Shiotani et al., 2007). The differences between the recovery times noted by our group and other investigators can be attributable the behavioral assessment scale used in the two studies. This study employed a 4 point scale which allows for gradations of recovery of function, Shiotani et al. used a 2-point scale: either the movement of the injured fold equaled that of

the uninjured side (recovery) or it did not (not recovered). The results of the current study demonstrate that treatment with supraphysiological levels of TP shortens the amount of time required to recover from vocal fold paralysis. The treatment with TP significantly decreased by two weeks the interval until full recovery of VFM from the RLN crush was achieved. This equates to a 33% reduction in overall recovery time. We hypothesize that this is due to TP changing the rate of regeneration as seen in the facial nerve injury and sciatic nerve injury paradigms (Brown et al., 1999, 2001; Hetzler et al., 2008; Jones, 1993; Jones & Oblinger, 1994; Kujawa et al., 1989, 1993; Sharma et al., 2009, 2010a, 2010b).

In considering the acceleration of behavioral recovery by 2 weeks in TP-treated animals, it is important to remember the relative distances that the RLN motoneurons must regenerate their axons in humans versus rats. To achieve recovery of function in the rat, axonal regeneration over millimeters must be accomplished. In humans, that distance increases to centimeters. Although the distances that motoneurons of the RLN must transport proteins needed for axonal regeneration are quite different between humans and rats, the rate at which the nerves themselves regenerate is not. Rates of peripheral nerve regeneration for rats have been reported in the range of 3.8 to 4.5 mm/day (Kujawa et al., 1993; Wang et al., 1997), while those for humans are somewhat lower, 0.6 – 1.9 mm/day (Sunderland, 1947; Polydefkis et al., 2004). The two factors, regeneration rate and relative distance, result in significantly longer times required for humans to regenerate peripheral nerves. Treatment with androgens in rats and hamsters has been shown to increase the rate of regeneration by 10–30% Jones, 1993; Jones & Oblinger, 1994; Kujawa et al., 1989, 1993. Given the length of nerve required to regenerate, a similar increase in rate in humans could greatly reduce the amount of time spent recovering from peripheral nerve injury not by weeks, but by months.

One potential pitfall of androgen treatment for peripheral nerve injury is the risk of accelerating the growth of androgen-dependent neoplasms in tissues that contain AR and are responsive to circulating levels of androgen (Schneider et al., 2007; Voelter et al., 2008). Whether or not laryngeal carcinomas would be affected by androgen treatment is still a matter of debate. While the results of some studies indicate that they appear to lack androgen receptors (Hagedorn et al., 2002; Bianchini et al., 2007), there may be co-incident expression of AR mRNA (Bianchini et al., 2007). However, since the mode of steroid delivery utilized in this study does increase systemic levels of androgen, this is still a valid concern. However, the beneficial effects of androgen on axonal regeneration appear to occur early in the regenerative response of the neuron. Given that, a dosing strategy which produces supraphysiological levels of androgen but can be terminated after the window of steroid effectiveness has passed would be most effective in stimulating axonal regeneration but with limited effects on the growth of hormone-dependent neoplasms. In both the current study as well as previous studies involving treatment with TP no deleterious effects were observed in the treated animals, such as increased autotomy, failure to thrive, increased aggression or other adverse behavioral effects.

Care should also be used in interpreting behavioral data derived from observations of VFM as an indicator of recovery of function and nerve regeneration. Recovery of function may not simply depend on the reinnervation of the laryngeal musculature by the RLN. Other investigators have determined that in the rat the abductor of the vocal fold, the posterior cricoarytenoid muscle (PCA), receives dual innervation from both the RLN and the superior laryngeal nerve (SLN) (Hydman & Mattsson, 2008). Recovery of function may in fact depend on collateral sprouting of the SLN as well as regeneration of the RLN (Hydman & MAttsson, 2008). As yet, no studies have investigated the effects of androgen treatment on SLN sprouting after denervation of the PCA after RLN injury. In an effort to compare the effects of treatment between that observed in experimental studies versus that seen in the clinics, a future direction may be to expand the outcome measure to compare the effects of androgen treatment after RLN injury on rat ultrasonic vocalizations.

5. Conclusion

This study is the first to demonstrate that androgens enhance the regeneration of the RLN after a crush injury as shown by the acceleration of recovery of complete VFM in some animals by 2 weeks or by a 33% decrease in overall recovery time. Additionally, the results show that the effects of androgens occur early in the regeneration process since the VFM was significantly improved by 1 and 2 weeks with TP. These findings are important as they provide further evidence

that the administration of supraphysiological concentrations of exogenous androgen immediately following injury has the potential as a therapy to be used clinically as well as extend the application of androgen as a neurotherapeutic agent beyond the facial nerve to other peripheral nerve injuries models. Overall, this study is the first of its kind to establish a non-surgical therapeutic approach to enhance regeneration and therefore functional recovery following a RLN injury.

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